Probability and Statistics

Chapter 1. Introduction to Statistics and Data Analysis

Jin Hee Yoon

Spring, 2026

Dept. of Artificial Intelligence and Information Technology
Sejong University

Chapter Overview

Why Statistics?

Sampling and Data Collection

Types of Variables

Frequencies and Graphs

Measures of Location

Measures of Variability

Summary

Why Statistics?

A Day in the Life of a Student

Consider a day in the life of a student.

You wake up in the morning and sunlight hits your eyes. Then, suddenly and without warning, the world becomes an uncertain place.

- How long will you have to wait for the Number 10 bus?
- When it arrives, will it be full?
- Will it be out of service?
- Will it be raining while you wait?
- Will you be late for your 9am mathematics lecture?

All of these questions involve **uncertainty**. Probability and statistics give us tools to study uncertainty using data.

Probability as the Science of Uncertainty

Probability is often called the **science of uncertainty**.

It is used in many fields:

- economics,
- finance,
- psychology,
- physics,
- engineering,
- data science and artificial intelligence.

Whenever outcomes are not known in advance, probability provides a mathematical language for modeling them.

Descriptive vs. Inferential Statistics

Descriptive Statistics

Methods for organizing, displaying, and describing data using

- tables,
- graphs,
- summary statistics.

Inferential Statistics

Methods for making decisions or predictions about a **population** based on **sampled data**.

- Descriptive statistics summarize what we observed.
- Inferential statistics go beyond the observed data.

Probability and Statistical Inference

$$\text{Population} \xrightarrow{\text{Probability}} \text{Sample}$$

$$\text{Sample} \xrightarrow{\text{Statistical Inference}} \text{Population}$$

- **Probability**: if the population is known, what kind of samples might we see?
- **Inference**: if a sample is observed, what can we learn about the population?

Sampling and Data Collection

Basic Definitions

Experimental unit: the individual or object on which a variable is measured.

Population: the complete set of items sharing a property of interest.

Sample: a set of data collected from a population by a defined procedure.

Data: the observed measurements; these may describe either a sample or an entire population.

Example 1: Student Survey

Suppose we study **hair color** among university students.

- **Variable**: hair color
- **Population**: all students at the university
- **Sample**: selected students obtained by a defined procedure
- **Experimental unit**: one student
- **Typical measurements**: brown, black, blonde, etc.

Example 2: Light Bulb Lifetime

Suppose the variable of interest is the **time until a light bulb burns out**.

- **Variable**: lifetime
- **Experimental unit**: one light bulb
- **Typical measurements**: 1500 hours, 1535.5 hours, etc.

Different experimental units may produce different values of the same variable.

How Many Variables Are Measured?

- **Univariate data**: one variable measured on a single experimental unit
- **Bivariate data**: two variables measured on a single experimental unit
- **Multivariate data**: more than two variables measured on a single experimental unit

Examples:

- Height only $\rightarrow$ univariate
- Height and weight $\rightarrow$ bivariate
- Height, weight, and blood pressure $\rightarrow$ multivariate

Types of Variables

Types of Variables

Variables

↓

Qualitative Quantitative

↓

Discrete Continuous

- **Qualitative variables** describe categories or characteristics.
- **Quantitative variables** measure numerical quantities.

Qualitative Variables

Qualitative variables measure a **quality** or **characteristic**.

Examples:

- hair color,
- make of car,
- gender,
- place of birth.

Quantitative Variables

Quantitative variables measure a **numerical quantity**.

- **Discrete**: a finite or countable number of possible values
- **Continuous**: infinitely many possible values on an interval

Examples:

- number of oranges on a tree,
- number of cars entering a campus,
- time until a light bulb burns out.

Frequencies and Graphs

Frequency, Relative Frequency, and Percent

A data distribution describes

- what values were observed,
- how often each value occurred.

Three common ways to measure "how often":

$$\text{Frequency}$$

$$\text{Relative Frequency} = \frac{\text{Frequency}}{n}$$

$$\text{Percent} = 100 \times \text{Relative Frequency}$$

Example: M&M Candies

Suppose a bag contains 25 candies.

Color	Tally	Frequency	Relative Frequency	Percent
Red	m m m m m	5	5/25 = .20	20%
Blue	m m m	3	3/25 = .12	12%
Green	m m	2	2/25 = .08	8%
Orange	m m m	3	3/25 = .12	12%
Brown	m m m m m m m m	8	8/25 = .32	32%
Yellow	m m m m	4	4/25 = .16	16%

Example: M&M Candies

If the number of red candies is 5, then:

$$\text{Frequency} = 5$$

$$\text{Relative Frequency} = \frac{5}{25} = 0.20$$

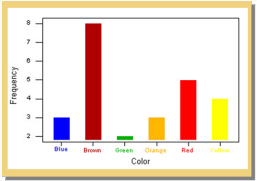
$$\text{Percent} = 20\%$$

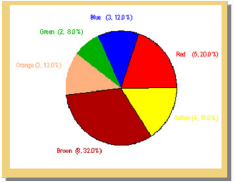
Frequency tells us the count. Relative frequency and percent tell us the proportion.

Graphs for Qualitative Data

Common displays for qualitative variables:

- **Bar chart**
- **Pie chart**





A bar chart is usually better for comparing categories, while a pie chart emphasizes proportions of the whole.

Stem-and-Leaf Plot

A stem-and-leaf plot is a simple graph for **quantitative data**.

- Divide each measurement into a **stem** and a **leaf**.
- List stems in a column.
- Record the corresponding leaves in the same row.
- This preserves much of the original numerical information.

Stem-and-Leaf Plot Example

Data:

$$90,\ 70,\ 70,\ 70,\ 75,\ 70,\ 65,\ 68,\ 60,\ 74,\ 70,\ 95,\ 75,\ 70,\ 68,\ 65,\ 40,\ 65$$

One possible stem-and-leaf display:

4	0
5	
6	0 5 5 5 8 8
7	0 0 0 0 0 0 4 5 5
8	
9	0 5

Stem-and-Leaf Plot Example

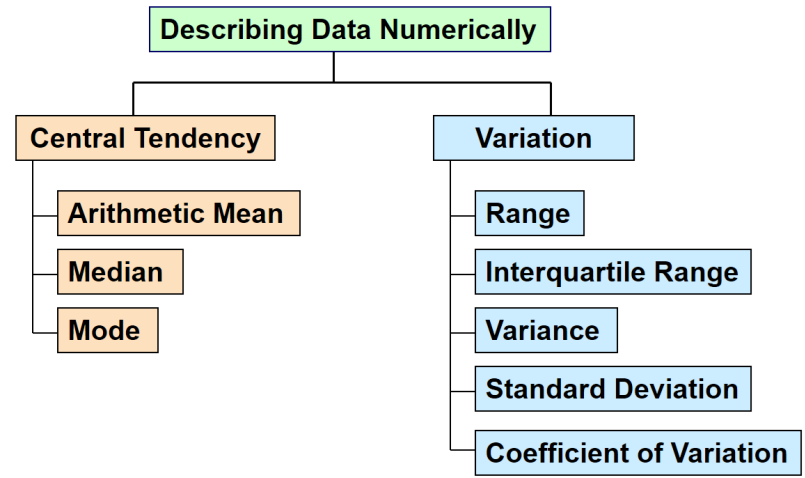
90	70	70	70	75	70	65	68	60
74	70	95	75	70	68	65	40	65

4	0
5	
6	5 8 0 8 5 5
7	0 0 0 5 0 4 0 5 0
8	
9	0 5

Reorder

4	0
5	
6	0 5 5 5 8 8
7	0 0 0 0 0 0 4 5 5
8	
9	0 5

Describing Data Numerically



Measures of Location

Measures of Central Tendency

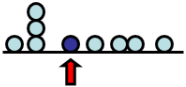
Overview

Central Tendency

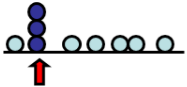
- **Mean**
- **Median**
- **Mode**

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Arithmetic average



Midpoint of ranked values



Most frequently observed value

Measures of Central Tendency

Three common measures of center:

- **Mean**
- **Median**
- **Mode**

These measures describe where the data are centered.

The Mean

For sample data $x_1, x_2, \ldots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

For a population of size N, the population mean is

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$$

- The mean is the most common measure of center.
- It is affected by extreme values.

The Median

The median is the **middle value** of an ordered data set.

$$\tilde{x} = \begin{cases} x_{\left(\frac{n+1}{2}\right)}, & \text{if } n \text{ is odd,} \\ \dfrac{x_{(n/2)} + x_{(n/2+1)}}{2}, & \text{if } n \text{ is even.} \end{cases}$$

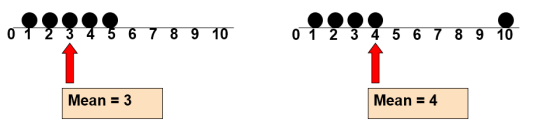
- 50% of the data lie below the median.
- 50% lie above the median.
- The median is less sensitive to extreme values.

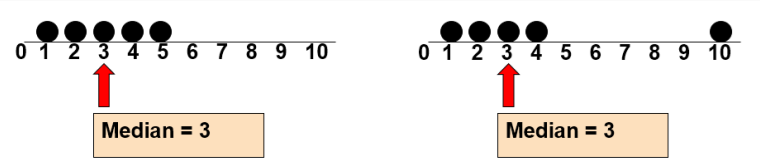
The Mode

The mode is the value that occurs **most often**.

- It can be used for numerical data.
- It can also be used for categorical data.
- A data set may have no mode.
- A data set may have one mode or several modes.

Simple Example

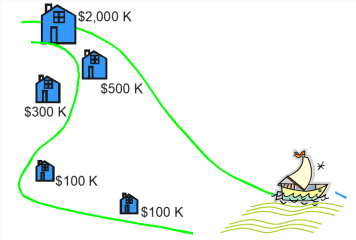




House Price Example

Consider five house prices:

$$2{,}000{,}000,\ 500{,}000,\ 300{,}000,\ 100{,}000,\ 100{,}000$$



$$\text{Mean} = \frac{3{,}000{,}000}{5} = 600{,}000$$

$$\text{Median} = 300{,}000$$

$$\text{Mode} = 100{,}000$$

A very large value pulls the mean upward, while the median remains more stable.

Which Measure Is Better?

- The **mean** is often used when the data are reasonably balanced.
- The **median** is often preferred when strong outliers are present.
- The **mode** is useful when the most common category or value matters.

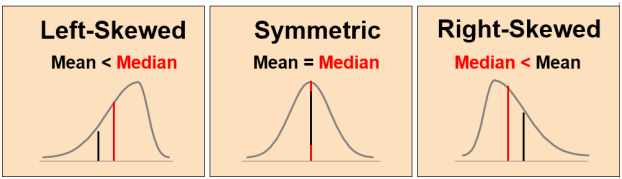
Shape of a Distribution

The relative positions of the mean and median help describe skewness.

Left-skewed: Mean < Median

Symmetric: Mean = Median

Right-skewed: Median < Mean



A long tail tends to pull the mean in its direction.

Measures of Variability

Why Variability Matters

Two data sets may have the same center but different spreads.

Common measures of variability:

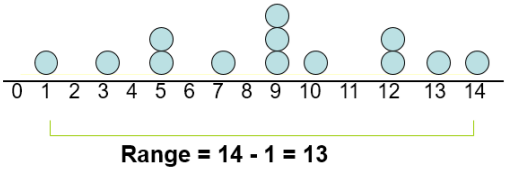
- **Range**
- **Qurartile**
- **Interquartile range (IQR)**
- **Variance**
- **Standard deviation**

Range

$$\text{Range} = x_{\max} - x_{\min}$$

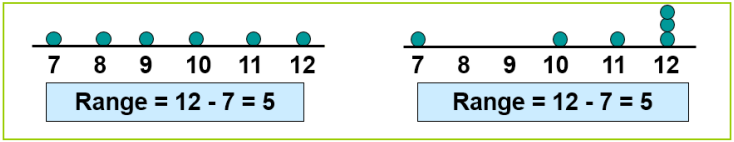
- The range depends only on the smallest and largest values.

Example:



Disadvantages of the Range

- Ignores the way in which data are distributed



- Sensitive to outliers

1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,3,3,3,3,4,5

Range = 5 - 1 = 4

1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,3,3,3,3,4,120

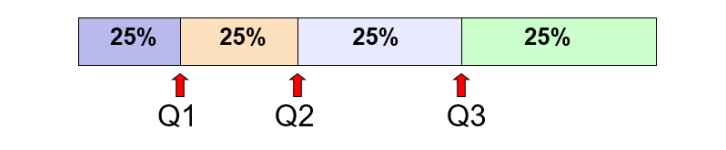
Range = 120 - 1 = 119

Quartiles

Quartiles divide ordered data into four equal parts.

- **First Quartile (Q_1)** 25% of the data lie below this value.
- **Second Quartile (Q_2)** This is the **median** of the data.
- **Third Quartile (Q_3)** 75% of the data lie below this value.

Quartiles describe the **location of data** within a distribution.



Quartile Formulas

To find quartiles, first arrange the data in **ascending order**.

Quartile positions in ranked data:

$$Q_1 = 0.25(n + 1)$$

$$Q_2 = 0.50(n + 1)$$

$$Q_3 = 0.75(n + 1)$$

where n is the number of observations

Example: Finding Quartiles

Ordered data:

$$4,\ 6,\ 7,\ 8,\ 10,\ 12,\ 13,\ 15$$

$$n = 8$$

Quartile positions:

$$Q_1 = 0.25(8+1) = 2.25$$

$$Q_2 = 0.50(8+1) = 4.5$$

$$Q_3 = 0.75(8+1) = 6.75$$

Quartiles are obtained by interpolating between the nearest ranked values.

Remarks for Quartiles

Quartiles divide an ordered data set into four parts.

For sorted data:

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$$

the quartiles are commonly denoted by

$$Q_1, \quad Q_2, \quad Q_3$$

- Q_1: first quartile, Q_2: second quartile, i.e. the median, Q_3: third quartile

Interpretation:

- about 25% of the data lie below Q_1,
- about 50% lie below Q_2,
- about 75% lie below Q_3.

Why Are There Different Quartile Definitions?

Quartiles seem simple, but there is **more than one valid definition**.

This happens because:

- data are often discrete and finite,
- the quartile position may not be an integer,
- some methods interpolate between observations,
- some methods choose actual observed values,
- software packages use different conventions.

Important Remark
Different textbooks and software may produce slightly different quartiles for the same data set.

Method 1: Median-of-Halves (Tukey-Style Idea)

One common classroom method is:

1. Sort the data.
2. Find the median Q_2.
3. Split the data into a lower half and an upper half.
4. Find the median of the lower half to get Q_1.
5. Find the median of the upper half to get Q_3.

Key Feature
This method is simple and intuitive, and it is often used in introductory statistics.

However, there is still one choice to make:

- If n is odd, do we include the overall median in each half or not?

That choice leads to slightly different versions.

Median-of-Halves: Excluding the Overall Median

Suppose the ordered data are

$$1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9$$

Then

$$Q_2 = 5$$

If we **exclude** the overall median, then:

$$\text{lower half: } 1,\ 2,\ 3,\ 4$$
$$\text{upper half: } 6,\ 7,\ 8,\ 9$$

So,

$$Q_1 = \frac{2+3}{2} = 2.5, \qquad Q_3 = \frac{7+8}{2} = 7.5$$

Median-of-Halves: Including the Overall Median

Using the same ordered data

$$1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9$$

we still have

$$Q_2 = 5$$

If we **include** the overall median in both halves, then:

$$\text{lower half: } 1,\ 2,\ 3,\ 4,\ 5$$
$$\text{upper half: } 5,\ 6,\ 7,\ 8,\ 9$$

Hence,

$$Q_1 = 3, \qquad Q_3 = 7$$

Method 2: Position Formula Method

Another common method uses quartile positions:

$$Q_1 \text{ at position } \frac{n+1}{4}$$
$$Q_2 \text{ at position } \frac{n+1}{2}$$
$$Q_3 \text{ at position } \frac{3(n+1)}{4}$$

Procedure:

- Sort the data.
- Compute the position of each quartile.
- If the position is an integer, take that observation.
- If not, interpolate between the two nearest observations.

Example: Position Formula Method

Consider the ordered data:

$$4,\ 6,\ 7,\ 8,\ 10,\ 12,\ 13,\ 15$$

Then $n = 8$.

Quartile positions are:

$$Q_1 : \frac{8+1}{4} = 2.25, Q_2 : \frac{8+1}{2} = 4.5, Q_3 : \frac{3(8+1)}{4} = 6.75$$

Thus:

- Q_1 lies between the 2nd and 3rd observations,
- Q_2 lies between the 4th and 5th observations,
- Q_3 lies between the 6th and 7th observations.

Interpolating Between Ranked Values

Using the previous data:

$$4,\ 6,\ 7,\ 8,\ 10,\ 12,\ 13,\ 15$$

$$Q_1 \text{ position } = 2.25$$

This means Q_1 is one-quarter of the way from $x_{(2)} = 6$ to $x_{(3)} = 7$:

$$Q_1 = 6 + 0.25(7 - 6) = 6.25$$

$$Q_2 = \frac{8 + 10}{2} = 9$$

$$Q_3 = 12 + 0.75(13 - 12) = 12.75$$

So the quartiles are

$$Q_1 = 6.25, \quad Q_2 = 9, \quad Q_3 = 12.75$$

Method 3: Software-Based Quantiles

In practice, software often uses quantile rules based on interpolation.

Examples:

- R provides several quantile types.
- Python libraries may use different interpolation rules.
- Spreadsheet software may use still another convention.

Typical idea:

- choose a percentile level p,
- compute a fractional rank,
- interpolate in a systematic way.

Important
There is no single universal software definition of quartiles.

A Simple Comparison of Quartile Methods

Consider the ordered data:

$$1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9$$

Possible results:

Method	Q_1	Q_2	Q_3
Median-of-halves (exclude median)	2.5	5	7.5
Median-of-halves (include median)	3	5	7
Position formula / interpolation	may differ	5	may differ

Conclusion
The quartiles depend on the definition being used.

Which Quartile Definition Should We Use?

For an introductory statistics course, a good strategy is:

- choose **one definition** clearly,
- use it consistently throughout the course,
- tell students that other books or software may differ slightly.

Common classroom choices:

- median-of-halves method,
- position formula method

$$Q_1 = \frac{n+1}{4}, \quad Q_2 = \frac{n+1}{2}, \quad Q_3 = \frac{3(n+1)}{4}.$$

Teaching Tip
What matters most is not memorizing every definition, but understanding that quartiles summarize the location of ordered data.

Interquartile Range (IQR)

Once quartiles are defined, we can compute the interquartile range:

$$IQR = Q_3 - Q_1$$

- The IQR measures the spread of the middle 50% of the data.
- It is less sensitive to outliers than the range.
- It depends on the quartile definition being used.

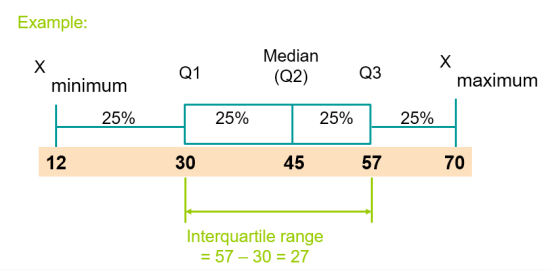
Remark
If two quartile definitions give different Q_1 and Q_3, then they may also give different IQR values.

Interquartile Range (IQR)

The **Interquartile Range** measures the spread of the middle 50% of the data.

$$IQR = Q_3 - Q_1$$

- IQR is resistant to extreme values.
- It focuses on the central portion of the data.
- It is widely used in boxplots.



Box Plot

A **box plot** is a graphical summary of a data set.

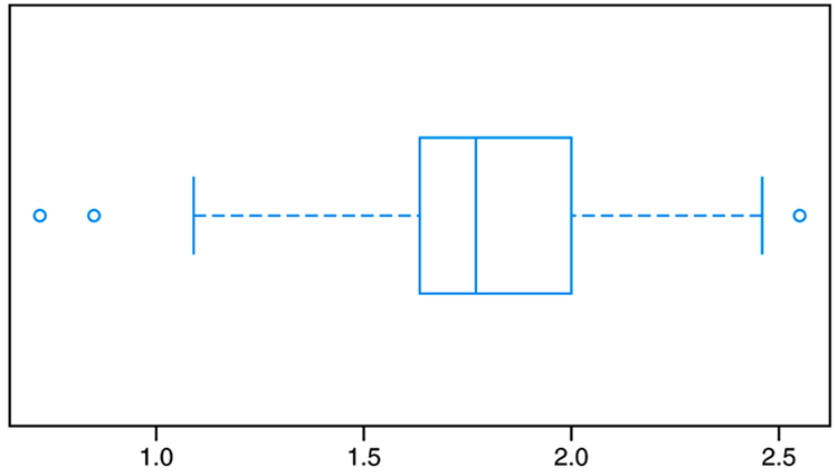
It is based on the **five-number summary**:

$$\text{minimum}, \quad Q_1, \quad \text{median}, \quad Q_3, \quad \text{maximum}$$

A box plot is useful because it shows:

- the center of the data,
- the spread of the data,
- possible skewness,
- possible outliers.

Example: Box Plot



Five-Number Summary

The five-number summary consists of:

- **Minimum**: the smallest observation (excluding outliers)
- **First quartile** Q_1: the 25th percentile
- **Median**: the 50th percentile
- **Third quartile** Q_3: the 75th percentile
- **Maximum**: the largest observation (excluding outliers)

These five values form the foundation of the box plot.

Outlier Rule in a Box Plot

A common rule labels an observation as an outlier if it lies outside

$$Q_1 - 1.5\,IQR \qquad \text{or} \qquad Q_3 + 1.5\,IQR$$

These values are called the **lower fence** and **upper fence**:

$$\text{Lower fence} = Q_1 - 1.5\,IQR$$

$$\text{Upper fence} = Q_3 + 1.5\,IQR$$

Observations outside these fences are plotted separately as outliers.

Main Parts of a Box Plot

A box plot has two main visual components:

- A **box** from Q_1 to Q_3
- A line inside the box showing the **median**

In addition, two **whiskers** extend outward from the box.

- The left whisker extends toward the lower end of the data
- The right whisker extends toward the upper end of the data

So the box plot gives a compact picture of location and spread.

Interquartile Range and the Box

The width of the box is determined by the **interquartile range (IQR)**:

$$IQR = Q_3 - Q_1$$

Interpretation:

- The box contains the middle 50% of the data.
- A wider box means greater variability in the middle half.
- A narrower box means less variability in the middle half.

Because it focuses on the middle 50%, the IQR is less sensitive to outliers than the range.

Whiskers in a Box Plot

The whiskers extend from the box toward the smallest and largest **non-outlying** observations.

Thus, the whiskers do not always reach the absolute minimum and maximum.

Their lengths help us understand:

- how far the data extend beyond the quartiles,
- whether the lower and upper tails are balanced,
- whether the distribution may be skewed.

How to Interpret a Box Plot

A box plot can reveal several important features.

- **Center**: shown by the median
- **Spread**: shown by the IQR and whiskers
- **Skewness**: suggested by asymmetry in the box or whiskers
- **Outliers**: shown as isolated points

A box plot does not show every individual observation, but it gives a quick and useful summary of the data.

Skewness in a Box Plot

A box plot may suggest the shape of the distribution.

- If the median is near the center of the box and the whiskers are similar, the distribution may be roughly symmetric.
- If the median is closer to Q_1 and the right whisker is longer, the distribution may be right-skewed.
- If the median is closer to Q_3 and the left whisker is longer, the distribution may be left-skewed.

Why Use a Box Plot?

A box plot is especially useful when we want to:

- summarize a data set quickly,
- compare several groups,
- detect unusual observations,
- examine spread and skewness.

It is widely used in exploratory data analysis.

Summary of Box Plot

- A box plot is based on the five-number summary.
- The box extends from Q_1 to Q_3.
- The line inside the box represents the median.
- The whiskers extend to the most extreme non-outlying observations.
- Outliers are often defined using the $1.5\,IQR$ rule.

A box plot provides a simple visual summary of center, spread, and possible outliers.

Population Variance

- Average of squared deviations of values from the mean

Population variance is defined as

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

Average squared distance of observations from the population mean

Population Variance: Notation

- μ = population mean
- N = population size
- x_i = i^{th} value of the variable x

Interpretation
Population variance measures how widely the population values are spread around the mean.

Sample Variance

The **sample variance** measures the average (approximately) of squared deviations from the sample mean.

For sample data $x_1, x_2, \ldots, x_n$,

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

where

- $\bar{x}$: arithmetic mean of the sample
- n: sample size
- x_i: i-th observed value

Population Standard Deviation

The **population standard deviation** is one of the most commonly used measures of variation.

It

- shows variation about the mean,
- has the same units as the original data.

Its formula is

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

where

- μ: population mean

Sample Standard Deviation

The **sample standard deviation** is also a very common measure of variation.

It

- shows variation about the sample mean,
- has the same units as the original data.

Its formula is

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Equivalently,

$$s = \sqrt{s^2}$$

Calculation Example: Sample Standard Deviation

Sample data:

$$10,\ 12,\ 14,\ 15,\ 17,\ 18,\ 18,\ 24$$

$$n = 8, \qquad \bar{x} = 16$$

First compute the sum of squared deviations:

$$(10-16)^2 + (12-16)^2 + (14-16)^2 + (15-16)^2$$
$$+(17-16)^2 + (18-16)^2 + (18-16)^2 + (24-16)^2 = 126$$

Then

$$s = \sqrt{\frac{126}{8-1}} = \sqrt{18} \approx 4.2426$$

Interpretation of Standard Deviation

The standard deviation is a measure of the **average scatter around the mean**.

- A **small standard deviation** means the data are concentrated close to the mean.
- A **large standard deviation** means the data are spread farther away from the mean.

Measuring Variation
Small standard deviation $\rightarrow$ narrow spread

Large standard deviation $\rightarrow$ wide spread

Center and Spread Should Be Read Together

- A single number cannot fully describe a data set.
- We should consider both **location** and **variability**.
- Graphical displays and numerical summaries should be used together.

Summary

Summary

- Probability studies uncertainty.
- Statistics organizes, summarizes, and interprets data.
- Inferential statistics uses sample data to learn about a population.
- Important concepts include population, sample, variable, and experimental unit.
- Variables may be qualitative or quantitative.
- Common measures of center are mean, median, and mode.
- Common measures of spread are range, IQR, variance, and standard deviation.